

DYNAMICAL SYSTEMS - MA3081

Solutions to Exercise Sheet 8

Exercise 8.1 (Normal forms).

Show that a system of the form

$$x' = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} x + \mathcal{O}(\|x\|^2)$$

can be locally transformed into the Poincaré normal form

$$y' = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} y + \begin{pmatrix} 0 \\ p_1 y_1^2 + p_2 y_1 y_2 \end{pmatrix} + \mathcal{O}(\|y\|^3).$$

Solution. We proceed as explained in the proof of Theorem 9.4, this time with $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$. Here we obtain (for the operator L given by $Lh^{(k)}(y) := Dh^{(k)}Ay - Ah^{(k)}(y)$)

$$M^{(2)} = \begin{pmatrix} 0 & 0 & 0 & -1 & 0 & 0 \\ 2 & 0 & 0 & 0 & -1 & 0 \\ 0 & 1 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Observe that $\text{rank}(M^{(2)}) = 4$ (third column is 0, second and last columns are linearly dependent), so we need a two-dimensional complementary space of resonant terms. The desired normal

$$y' = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} y + \begin{pmatrix} 0 \\ p_1 y_1^2 + p_2 y_1 y_2 \end{pmatrix} + \mathcal{O}(\|y\|^3).$$

already gives us a suggestion for the complementary space, since we have resonant terms that are linear combination of $\begin{pmatrix} 0 \\ y_1^2 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ y_1 y_2 \end{pmatrix}$.

It is easy to check that the vectors $(0, 0, 0, 1, 0, 0)^\top$ and $(0, 0, 0, 0, 1, 0)^\top$ indeed complement the range of $M^{(2)}$.

Remark: Note that the resonant terms do not need to be chosen in the way of Lemma 9.3. This lemma only gives one (or two) possible choices for a space of resonant terms. Any complementary space to $L(\mathcal{H}_k)$ (i.e. a subspace $\mathcal{R}_k$ such that $L(\mathcal{H}_k) \oplus \mathcal{R}_k = \mathcal{H}_k$) is eligible. $\square$

Exercise 8.2 (Normal form theory for maps).

- a) Consider the dynamical system induced by iteration of a map $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$. Transfer the considerations of Section 9 to this setting of a discrete-time dynamical system. Specifically, determine the form of the homological equation.

b) Apply a) to eliminate as many quadratic terms as possible for the map

$$f: \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \begin{pmatrix} x + (x - y)^2 \\ 3y + x^2 + 2y^2 \end{pmatrix}.$$

Solution.

a) We start with some mapping

$$x \mapsto f(x) = Ax + \sum_{j=2}^k f^{(j)}(x) + \mathcal{O}(x^{k+1}). \quad (8.1)$$

Also for a discrete time dynamical system we introduce new coordinates y given by

$$x = H(y) = y + h^{(k)}(y),$$

where $h^{(k)} \in \mathcal{H}_k$ is a homogeneous polynomial of degree k . The inverse of H is given by

$$y = H^{-1}(x) = x - h^{(k)}(x) + \mathcal{O}(x^{k+1}).$$

This transforms (8.1) into

$$\begin{aligned} y &\mapsto H^{-1} \circ f \circ H(y) \\ &= (id - h^{(k)}(\cdot) + \mathcal{O}((\cdot)^{k+1})) \left(A(y + h^{(k)}(y)) + \sum_{j=2}^k f^{(j)}(y + h^{(k)}(y)) + \mathcal{O}((y + h^{(k)}(y))^{k+1}) \right) \\ &= (id - h^{(k)}(\cdot) + \mathcal{O}((\cdot)^{k+1})) \left(Ay + Ah^{(k)}(y) + \sum_{j=2}^k f^{(j)}(y) + \mathcal{O}(y^{k+1}) \right) \\ &= Ay + Ah^{(k)}(y) + \sum_{j=2}^k f^{(j)}(y) - h^{(k)}(Ay) + \mathcal{O}(y^{k+1}) \end{aligned}$$

Hence we define $L : \mathcal{H}_k \rightarrow \mathcal{H}_k$ by $(Lh^{(k)})(y) = h^{(k)}(Ay) - Ah^{(k)}(y)$ which yields the homological equation

$$(Lh^{(k)})(y) = h^{(k)}(Ay) - Ah^{(k)}(y) = f^{(k)}(y).$$

b) To see which terms can be eliminated we proceed again as in the proof of Theorem 9.4 or in Exercise 6.2, but now use the different definition of the operator L above.

We consider the map

$$\begin{aligned} f : \mathbb{R}^2 &\rightarrow \mathbb{R}^2 \\ \begin{pmatrix} x \\ y \end{pmatrix} &\mapsto \begin{pmatrix} x + (x - y)^2 \\ 3y + x^2 + 2y^2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} x^2 - 2xy + y^2 \\ x^2 + 2y^2 \end{pmatrix}. \end{aligned}$$

We have

$$\begin{aligned} Lh \begin{pmatrix} x \\ y \end{pmatrix} &= h \left(A \begin{pmatrix} x \\ y \end{pmatrix} \right) - Ah \begin{pmatrix} x \\ y \end{pmatrix} \\ &= h \left[\begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \right] - \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} h_1(x, y) \\ h_2(x, y) \end{pmatrix} \\ &= h \begin{pmatrix} x \\ 3y \end{pmatrix} - \begin{pmatrix} h_1(x, y) \\ 3h_2(x, y) \end{pmatrix} \\ &= \begin{pmatrix} h_1(x, 3y) - h_1(x, y) \\ h_2(x, 3y) - 3h_2(x, y) \end{pmatrix}. \end{aligned}$$

Plugging in the basis elements, we find

$$L \begin{pmatrix} x^2 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad L \begin{pmatrix} xy \\ 0 \end{pmatrix} = \begin{pmatrix} 2xy \\ 0 \end{pmatrix}, \quad L \begin{pmatrix} y^2 \\ 0 \end{pmatrix} = \begin{pmatrix} 8y^2 \\ 0 \end{pmatrix}$$

for the first component, and for the second component

$$L \begin{pmatrix} 0 \\ x^2 \end{pmatrix} = \begin{pmatrix} 0 \\ -2x^2 \end{pmatrix}, \quad L \begin{pmatrix} 0 \\ xy \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad L \begin{pmatrix} 0 \\ y^2 \end{pmatrix} = \begin{pmatrix} 0 \\ 6y^2 \end{pmatrix}.$$

We want to identify L with some $M^{(2)} \in \mathbb{R}^{6 \times 6}$. Using the coordinate ordering in $\mathbb{R}^6$ given by

$$(x^2 \partial_x, xy \partial_x, y^2 \partial_x, x^2 \partial_y, xy \partial_y, y^2 \partial_y)$$

we find

$$M^{(2)} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 6 \end{pmatrix}.$$

We immediately see that a simple choice of a complementary space is given by

$$\text{span} \left\{ (1, 0, 0, 0, 0, 0)^\top, (0, 0, 0, 0, 1, 0)^\top \right\},$$

i.e. resonant terms are linear combinations of $\begin{pmatrix} x^2 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ xy \end{pmatrix}$.

Since our map f does not contain quadratic terms of the type $\begin{pmatrix} 0 \\ xy \end{pmatrix}$, only the term x^2 in the first component cannot be eliminated; hence a normal form is given by

$$f(x, y) = \begin{pmatrix} 1 & 0 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} + \begin{pmatrix} x^2 \\ 0 \end{pmatrix} + \mathcal{O}(\|(x, y)\|^3).$$

(If you choose a different space $\mathcal{R}_k$ of resonant terms, other normal forms might arise.)

□

Exercise 8.3 (Center manifold I).

Consider the dynamical system induced by

$$\begin{aligned} x' &= x^2 - xy, \\ y' &= x - y + x^2 - xy + xz, \\ z' &= -3z + x^2 \end{aligned}$$

around the equilibrium point at the origin.

- Find a linear coordinate transformation that brings the system into the form (10.3) with separated center and stable/unstable components.
- Find an approximation of the center manifold up to cubic order.

c) Determine the stability of the equilibrium at the origin.

Solution.

a) The linearization at the origin is given by the matrix $\begin{pmatrix} 0 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -3 \end{pmatrix}$ with eigenvalues

$\lambda_1 = 0, \lambda_2 = -1, \lambda_3 = -3$, so $d_c = 1$ and $d_s = 2$.

For the form of equation (10.3) we need to bring this matrix into diagonal form, so we use a coordinate transformation matrix T consisting of eigenvectors, which is given by

$$T = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \text{ The inverse thereof is } T^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

New coordinates u, v, w are now defined by $T \begin{pmatrix} u \\ v \\ w \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$.

In these coordinates the differential equations read

$$\begin{aligned} u' &= x' = x^2 - xy = u^2 - u(u+v) = -uv, \\ v' &= y' - x' = x - y + x^2 - xy + xz - x^2 + xy = x - y + xz = -v + uw, \\ w' &= z' = -3z + x^2 = -3w + u^2 \end{aligned}$$

The new equations

$$\begin{aligned} u' &= -uv, \\ v' &= -v + uw, \\ w' &= -3w + u^2 \end{aligned}$$

now have the desired form of (10.3) with $B = 0$ and $C = \begin{pmatrix} -1 & 0 \\ 0 & -3 \end{pmatrix}$.

b) The center direction is given by the u -coordinate, while (v, w) form the stable directions. Hence the center manifold $W^c(0)$ is given as the graph of some function $H : \mathbb{R} \rightarrow \mathbb{R}^2$ over the u -direction. We make the following ansatz for an approximation h to H :

$$\begin{aligned} v &= h_1(u) = a_1 u^2 + b_1 u^3, \\ w &= h_2(u) = a_2 u^2 + b_2 u^3. \end{aligned} \tag{8.2}$$

The invariance equation is given by

$$\begin{aligned} h_1'(u)[-uh_1(u)] - (-h_1(u) + uh_2(u)) &= 0, \\ h_2'(u)[-uh_1(u)] - (-3h_2(u) + u^2) &= 0. \end{aligned}$$

Inserting the ansatz (8.2) into these equations we obtain

$$\begin{aligned} a_1 u^2 + b_1 u^3 - a_2 u^3 + \mathcal{O}(u^4) &= 0, \\ 3a_2 u^2 + 3b_2 u^3 - u^2 + \mathcal{O}(u^4) &= 0, \end{aligned}$$

so we need to have

$$a_1 = 0, \quad b_1 - a_2 = 0, \quad 3a_2 - 1 = 0, \quad b_2 = 0$$

in order to solve the invariance equation up to $\mathcal{O}(u^4)$. This yields

$$a_1 = 0, \quad b_1 = \frac{1}{3}, \quad a_2 = \frac{1}{3}, \quad b_2 = 0$$

meaning the center manifold is given as the graph of $H(u) = \begin{pmatrix} \frac{1}{3}u^3 + \mathcal{O}(u^4) \\ \frac{1}{3}u^2 + \mathcal{O}(u^4) \end{pmatrix}$.

c) The reduced system on the center manifold reads

$$u' = -uH_1(u) = -\frac{1}{3}u^4 + \mathcal{O}(u^5) .$$

The stability of the origin of this one-dimensional equation can now be analyzed without relying on the linearization. This vector field is negative for all $u \neq 0$ in some small neighbourhood of 0 since the term $-\frac{1}{3}u^4$ dominates. Thus negative initial conditions sufficiently close to 0 are repelled from the origin, hence it is an unstable equilibrium.

Consequently also the equilibrium $(u, v, w) = (x, y, z) = (0, 0, 0)$ of our original system is unstable.

□

Exercise 8.4 (Center manifold II).

We consider the following system of ordinary differential equations

$$\begin{aligned} x' &= -y + xz + 2y^4, \\ y' &= x + yz + x^2y^2, \\ z' &= -z - (x^2 + y^2) + z^2. \end{aligned}$$

Find a quadratic approximation of the center manifold of the equilibrium point at the origin. Use that to determine the stability of the equilibrium and sketch the phase portrait near the origin.

Solution. Linearizing at the origin yields

$$\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

with eigenvalues $\lambda_{1,2} = \pm i$ and $\lambda_3 = -1$. Note that this is already the form of (10.3) with $B = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $C = -1$.

The center manifold is two-dimensional, given as a graph $\{z = H(x, y)\}$. Let us consider a quadratic polynomial ansatz $z = h(x, y) = ax^2 + bxy + cy^2$.

The invariance equation reads:

$$\partial_x h(-y + xh + 2y^4) + \partial_y h(x + yh + x^2y^2) - (-h - (x^2 + y^2) + h^2) = 0$$

Plugging in our ansatz yields

$$\begin{aligned} 0 &= (2ax + by)(-y + \mathcal{O}(3)) + (2cy + bx)(x + \mathcal{O}(3)) + (ax^2 + bxy + cy^2) + (x^2 + y^2) + \mathcal{O}(3) \\ &= -2axy - by^2 + 2cxy + bx^2 + ax^2 + bxy + cy^2 + x^2 + y^2 + \mathcal{O}(3) \\ &= (a + b + 1)x^2 + (-2a + b + 2c)xy + (-b + c + 1)y^2 + \mathcal{O}(3) \end{aligned}$$

with $\mathcal{O}(3) := \mathcal{O}(\|(x, y)\|^3)$.

Hence the coefficients must satisfy

$$\begin{aligned} a + b + 1 &= 0, \\ -2a + b + 2c &= 0, \\ -b + c + 1 &= 0, \end{aligned}$$

for that the invariance equation is solved up to order $\mathcal{O}(3)$. The solution of this system of linear equations is given by

$$a = -1, \quad b = 0, \quad c = -1,$$

consequently our approximation yields

$$H(x, y) = -(x^2 + y^2) + \mathcal{O}(3) .$$

Substituting this into the first two equations of the vector field we obtain

$$\begin{aligned} x' &= -y - x(x^2 + y^2) + \mathcal{O}(4) \\ y' &= x - y(x^2 + y^2) + \mathcal{O}(4) \end{aligned}$$

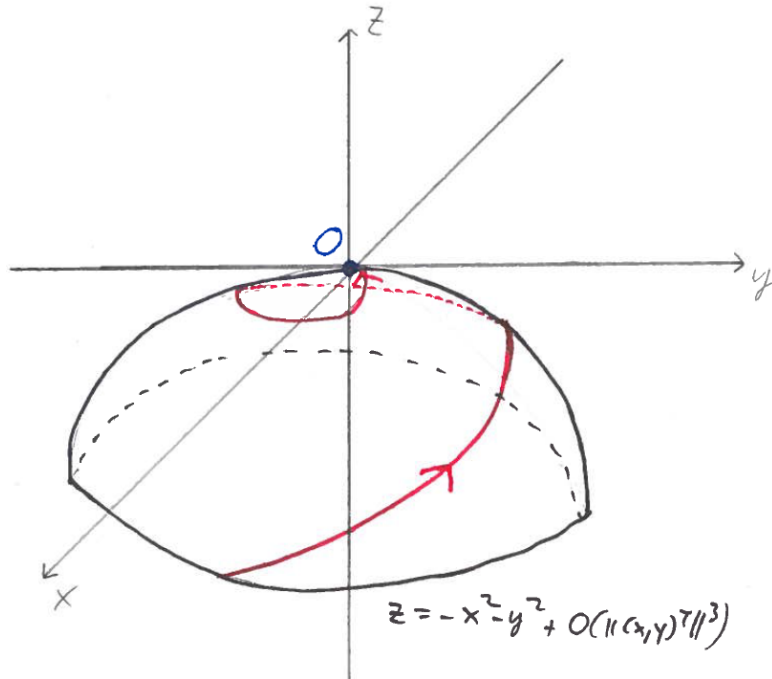
To analyse the stability of the equilibrium on the center manifold we can use the Lyapunov function $L(x, y) = x^2 + y^2$ or introduce polar coordinates. In the latter case, the reduced equation is transformed into

$$\begin{aligned} r' &= -r^3 + \mathcal{O}(r^4) \\ \theta' &= 1 + \mathcal{O}(r^3) \end{aligned}$$

In particular, since $r > 0$, we see that $r' < 0$ in a neighbourhood of $(x, y) = (0, 0)$ for the reduced system, so the equilibrium is stable.

Since for the original system $d_u = 0$, i.e. all remaining components (here z -component) are stable, we can conclude that the equilibrium $(x, y, z) = (0, 0, 0)$ is stable.

The phase portraits around the origin looks as follows (trajectories outside the center manifold will quickly converge towards the center manifold and then also start a spiraling behaviour towards the origin).



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